# The $t_c$-Cederbaum model: ghost Gutzwiller by direct minimization

(core fragment with $B_c = 1$, 3 and 5 quasiparticle orbitals)

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This note defines the Hamiltonian that is solved, the $t_c$-Cederbaum model (the Cederbaum model of core-hole satellites plus a weak core–virtual hopping $t_c$; $t_c = 0$ is the original Cederbaum model), its exact solution by diagonalization (ED), the ghost-Gutzwiller (gGut) variational problem used for it, the direct minimization that solves that problem, and the comparison of the gGut results with ED, for a core fragment without ghosts ($B_c = 1$) and with two and four ghosts ($B_c = 3$, 5). The aim of the test is to see whether gGut reproduces the satellite structure of the core-hole spectrum of this model. Quantities are per spin unless stated otherwise; solutions are real and paramagnetic.

## 1 The Hamiltonian

Three orbitals, $c$ (core), $o$ (occupied) and $v$ (virtual), each with two spins. The $t_c$-Cederbaum model is $H_{\mathrm{tot}} = H + H_t$, where $H$ is the Cederbaum model (the $o$–$v$ hopping conditioned on a core hole) and $H_t$ the added hopping:

$$\begin{aligned} H_{\mathrm{tot}} = {} & g \sum_{\sigma} (n_{v\sigma} - n_{o\sigma}) - E_c \sum_{\sigma} n_{c\sigma} - V \sum_{\sigma\sigma'} (1 - n_{c\sigma}) \left( c^{\dagger}_{v\sigma'} c_{o\sigma'} + \mathrm{h.c.} \right) \\ & + t_c \sum_{\sigma} \left( c^{\dagger}_{c\sigma} c_{v\sigma} + \mathrm{h.c.} \right), \end{aligned} \tag{1}$$

with $E_c > 0$, orbital gap $g$, core–valence coupling $V$, and a weak core–virtual hopping $t_c$. The prefactor $\sum_{\sigma}(1 - n_{c\sigma}) = 2 - n_c$ counts the core holes, so the $o$–$v$ hopping acts only when the core is not full. The particle number and $S_z$ are conserved. The core occupation $n_c$ is conserved only at $t_c = 0$. Energies are measured in units of $g$ ($g = 1$), so the parameters are the ratios $E_c/g$, $V/g$ and $t_c/g$. All results below use $E_c/g = 10$, $V/g \in \{1/\sqrt{2}, \sqrt{2}\} = \{0.707, 1.414\}$ and $t_c/g \leq 0.5$; the ground state then lies in the sector with two electrons of each spin. The two values of $V$ are chosen so that $\rho/\Omega = V/(\sqrt{2}\, g) = 0.5$ and 1.0, the ratios of the transition moment $\rho$ to the excitation energy $\Omega = 2g$ used by Kockläuner and Golze for this model [2] (Sec. 6.3); $E_c$ only shifts all poles rigidly at $t_c = 0$.

## 2 Exact diagonalization

**Diagonalization.** $H_{\mathrm{tot}}$ is diagonalized exactly in the 64-dimensional Fock space of six spin orbitals (Jordan–Wigner operators). At $t_c = 0$ the ground state is the product state $|220\rangle$ (notation $|n_c n_o n_v\rangle$: core and $o$ doubly occupied, $v$ empty) with $E = -2E_c - 2g$ and core occupation

$\nu_c = \langle n_c \rangle = 2$; for $t_c > 0$ it acquires a core-hole component and $\nu_c < 2$. The core-hole Green's function for spin $\uparrow$ is

$$G^h_{cc}(\omega) = \sum_n \frac{|\langle n|c_{c\uparrow}|\mathrm{GS}\rangle|^2}{\omega + (E_n - E_{\mathrm{GS}}) - i\eta}, \tag{2}$$

with poles at $\omega = -E_i$ and weights $p_i$. At $t_c = 0$ there are three poles; with $\alpha = V/g$,

$$\begin{aligned} E_1 &= E_c + 2g - 2g\sqrt{1+\alpha^2}, & p_1 &= \frac{(1+\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \\ E_2 &= E_c + 2g, & p_2 &= \frac{\alpha^2}{2(1+\alpha^2)}, \\ E_3 &= E_c + 2g + 2g\sqrt{1+\alpha^2}, & p_3 &= \frac{(1-\sqrt{1+\alpha^2})^2}{4(1+\alpha^2)}, \end{aligned} \tag{3}$$

which the ED reproduces to $9 \times 10^{-15}$. For $t_c > 0$ the poles move at $O(t_c^2)$ and the middle pole splits into two lines $\sim 10^{-4}$ apart (Tab. 2); Tab. 1 gives the ground-state quantities.

| $V/g$ | $t_c/g$ | $(E_0 + 2E_c + 2g)/g$ | $2 - \nu_c$ | $\rho^{cv} = \langle c^\dagger_{c\uparrow} c_{v\uparrow}\rangle$ |
|---|---|---|---|---|
| 0.707 | 0.5 | $-4.55 \times 10^{-2}$ | $4.15 \times 10^{-3}$ | $-4.54 \times 10^{-2}$ |
| 0.707 | 0.2 | $-7.30 \times 10^{-3}$ | $6.67 \times 10^{-4}$ | $-1.82 \times 10^{-2}$ |
| 0.707 | 0.1 | $-1.82 \times 10^{-3}$ | $1.67 \times 10^{-4}$ | $-9.12 \times 10^{-3}$ |
| 0.707 | 0.05 | $-4.56 \times 10^{-4}$ | $4.17 \times 10^{-5}$ | $-4.56 \times 10^{-3}$ |
| 1.414 | 0.5 | $-4.60 \times 10^{-2}$ | $4.27 \times 10^{-3}$ | $-4.59 \times 10^{-2}$ |
| 1.414 | 0.2 | $-7.37 \times 10^{-3}$ | $6.87 \times 10^{-4}$ | $-1.84 \times 10^{-2}$ |
| 1.414 | 0.1 | $-1.84 \times 10^{-3}$ | $1.72 \times 10^{-4}$ | $-9.22 \times 10^{-3}$ |
| 1.414 | 0.05 | $-4.61 \times 10^{-4}$ | $4.30 \times 10^{-5}$ | $-4.61 \times 10^{-3}$ |

Table 1: Exact ground state of $H_{\mathrm{tot}}$: $2 - \nu_c \propto t_c^2$ and $\rho^{cv} \propto t_c$.

| $V/g$ | | $-E_1$ | $-E_2$ (split pair) | $-E_3$ | $p_1$ |
|---|---|---|---|---|---|
| 0.707 | $t_c \to 0$ (Eq. 3) | −9.5505 | −12 | −14.4495 | 0.8249 |
| | $t_c/g = 0.5$ | −9.5777 | −12.0441, −12.0477 | −14.5121 | 0.8268 |
| | $t_c/g = 0.1$ | −9.5516 | −12.0018, −12.0019 | −14.4520 | 0.8250 |
| | $t_c/g = 0.05$ | −9.5508 | −12.0004, −12.0005 | −14.4501 | 0.8249 |
| 1.414 | $t_c \to 0$ (Eq. 3) | −8.5359 | −12 | −15.4641 | 0.6220 |
| | $t_c/g = 0.5$ | −8.5707 | −12.0406, −12.0510 | −15.5198 | 0.6263 |
| | $t_c/g = 0.1$ | −8.5373 | −12.0016, −12.0020 | −15.4663 | 0.6222 |
| | $t_c/g = 0.05$ | −8.5362 | −12.0004, −12.0005 | −15.4647 | 0.6221 |

Table 2: Exact hole poles of $G^h_{cc}$ at $V/g = 0.707$ and 1.414 (weights $> 10^{-3}$); the split pair has total weight $p_2 = \alpha^2/[2(1+\alpha^2)] = 0.1667$ and 0.3333, respectively.

## 3 The ghost-Gutzwiller variational problem

**Fragments and quasiparticle space.** Each orbital $I \in \{c, o, v\}$ is a fragment with $B_I$ quasiparticle orbitals $d_{Ia\sigma}$, $a = 1, \ldots, B_I$: $B_o = B_v = 1$ (no local interaction) and $B_c \in \{1, 3, 5\}$, i.e. $B_c - 1$ ghost orbitals on the core. The Gutzwiller wave function is $|\Psi_G\rangle = \prod_I P_I |\psi_{qp}\rangle$ with $|\psi_{qp}\rangle$

a Slater determinant that fills $N_{\rm occ} = 2 + (B_c - 1)/2$ of the $B_c + 2$ levels per spin. Its one-body density matrix is $\rho^{IJ}_{ab} = \langle d^\dagger_{Ia\sigma} d_{Jb\sigma}\rangle$.

**Embedding states.** The projector of each fragment is represented by a real normalized state $\Phi_I$ in the Fock space of the physical orbital $I$ plus $B_I$ bath orbitals per spin, restricted to the half-filled sector with $(1 + B_I)/2$ electrons of each spin ($B_I$ odd). Writing $c_{I\sigma}$ for the physical and $d_{Ia\sigma}$ for the bath orbitals, the following quantities are expectation values in $\Phi_I$ (spin averaged):

$$\begin{aligned} D_{I,a} &= \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} c_{I\sigma}\rangle, \quad \Delta^{bb}_{I,ab} = \tfrac{1}{2}\sum_\sigma \langle d^\dagger_{Ia\sigma} d_{Ib\sigma}\rangle, \\ \nu_c &= \langle n_c\rangle, \qquad\qquad\qquad K_{\gamma\delta} = \Big\langle n_c \sum_\sigma d_{c\delta\sigma} d^\dagger_{c\gamma\sigma}\Big\rangle, \end{aligned} \tag{4}$$

and $E^{loc}_I = \langle H^{loc}_I\rangle$ with $H^{loc}_c = -E_c n_c$, $H^{loc}_o = -g n_o$, $H^{loc}_v = +g n_v$ ($n_I = \sum_\sigma n_{I\sigma}$). From them:

$${}^I\Delta = \mathbb{I} - \Delta^{bb}_I, \qquad S_I = \big[{}^I\Delta(\mathbb{I} - {}^I\Delta) + \varepsilon\,\mathbb{I}\big]^{1/2}, \qquad R_I = S_I^{-1} D_I, \qquad C = (\mathbb{I} - {}^c\Delta + \varepsilon\,\mathbb{I})^{-1}, \tag{5}$$

where $\varepsilon$ is a small regularizing shift ($10^{-9}$ unless stated) and $R_I \in \mathbb{R}^{B_I}$ is the renormalization vector, $c^\dagger_{I\sigma} \to \sum_a R_{I,a} d^\dagger_{Ia\sigma}$. The non-local core–valence correlation enters through the tensor

$$\bar\Theta = \tfrac{1}{2} S_c^{-1} K S_c^{-1} - \nu_c\, C, \qquad u = \rho^{cv} R_v, \quad w = \rho^{co} R_o, \tag{6}$$

which follows from evaluating $\langle\Psi_G| n_{c\sigma} c^\dagger_{o\sigma'} c_{v\sigma'} |\Psi_G\rangle$ with the Gutzwiller approximation (derived in `main.tex`, Eqs. ThetaEff–ThetaEqn, and `SI.tex`, S2).

**Energy functional.** The energy is $E = \langle\Psi_G|H_{\rm tot}|\Psi_G\rangle$, evaluated term by term from Eq. (1) with $c^\dagger_{I\sigma} \to \sum_a R_{I,a} d^\dagger_{Ia\sigma}$ and Wick's theorem on $|\psi_{qp}\rangle$.

- *Local terms.* $g(n_v - n_o) - E_c n_c = \sum_I H^{loc}_I$ is evaluated in the embedding states, $\sum_I E^{loc}_I$.
- *Core–virtual hopping.* Per spin, $\langle c^\dagger_{c\sigma} c_{v\sigma}\rangle = R_c^T \rho^{cv} R_v$. The sum over the two spins and the hermitian conjugate give a factor 4: $4t_c\, R_c^T \rho^{cv} R_v$.
- *Conditioned o–v hopping.* The weight $(2 - n_c)$ multiplies $\sum_{\sigma'}(c^\dagger_{v\sigma'} c_{o\sigma'} + \text{h.c.})$ and the expectation value does not factorize, since $n_c$ fluctuates and is correlated with the hopping. Per spin, $\langle (2 - n_c)\, c^\dagger_{v\sigma} c_{o\sigma}\rangle = (2 - \nu_c)\, R_v^T \rho^{vo} R_o + w^T \bar\Theta\, u$: the first part is the mean-field one, with the average number of core holes $2 - \nu_c$; the second is the connected core–valence correlation, built from $\bar\Theta$, $u$ and $w$ defined above. Spin sum and hermitian conjugate again give a factor 4.

Adding the three contributions with the prefactors of Eq. (1):

$$E = \underbrace{\sum_I E^{loc}_I}_{\text{local}} + \underbrace{4t_c\, R_c^T \rho^{cv} R_v}_{t_c\ \text{hopping}} - \underbrace{4V(2 - \nu_c)\, R_v^T \rho^{vo} R_o}_{o\text{–}v\ \text{mean field}} - \underbrace{4V\, w^T \bar\Theta\, u}_{\text{connected}}. \tag{7}$$

Written out with the definitions above, the dependences are: $E^{loc}_I$, $\nu_c$, $\Delta^{bb}_I$ and $K$ are expectation values in $\Phi_I$; $R_I = S_I^{-1} D_I$ with $S_I$ a function of $\Delta^{bb}_I$; $\bar\Theta$ a function of $\nu_c$, $K$ and ${}^c\Delta$; $u = \rho^{cv} R_v$ and $w = \rho^{co} R_o$. So $E$ depends on the embedding states only through $E^{loc}_I$, $D_I$, $\Delta^{bb}_I$, $\nu_c$ and $K$, and on $|\psi_{qp}\rangle$ only through $\rho$.

**The limit $t_c = 0$.** At $t_c = 0$ the core occupation is conserved and the exact ground state is the product state $|220\rangle$ of Sec. 2, with $\nu_c = 2$. There is no core–valence correlation, so the core is coherent neither with $v$ nor with $o$: $\rho^{cv} = \rho^{co} = 0$ and $u = w = 0$. Term by term in Eq. (7):

- the $t_c$ hopping vanishes through its explicit factor $t_c$;
- the mean-field $o$–$v$ term vanishes through $2 - \nu_c = 0$;
- the connected term vanishes through $u = w = 0$.

Hence

$$E\big|_{t_c=0} = \sum_I E_I^{loc} = -2E_c - 2g, \tag{8}$$

which is the exact energy. Two consequences follow. First, $V$ does not appear: every $V$ term of Eq. (7) is zero. Second, $R_c = S_c^{-1} D_c$ appears in Eq. (7) *only* in the $t_c$ hopping term (the other quantities, $E_I^{loc}$, $\nu_c$, $\Delta_I^{bb}$, $K$, do not involve $D_c$, and $u$, $w$ contain $R_v$, $R_o$ but not $R_c$), so with that term gone $E$ is independent of $D_c$ and the minimization leaves $R_c$ undetermined. At $V = 0$ and $t_c \neq 0$ the situation is different: the two $V$ terms drop out, but $E = \sum_I E_I^{loc} + 4t_c\, R_c^T \rho^{cv} R_v$ still depends on $R_c$, which is therefore determined by the hopping.

These two consequences are the reason for the hopping $t_c$. The quantity of interest is not the ground-state energy but the core-hole Green's function $G^h_{cc}$, whose three poles are final-state physics: the three $N - 1$ eigenstates reached when a core electron is removed and $V$ then moves electrons between $o$ and $v$. gGut does not compute $N - 1$ states; the spectrum comes from the quasiparticle Hamiltonian $H_{qp}$ (Sec. 4), whose parameters $R_I$, $\lambda_I$ are fixed by the ground-state variational problem. At $t_c = 0$ that problem contains no $V$ and leaves $R_c$ free, so the $G^h_{cc}$ built from $R_c$ carries no information about the satellites, although the ground state itself is exact. A weak $t_c$ cures both defects at once: it makes $R_c$ determined and slightly correlates the ground state ($\nu_c < 2$), so that the $V$ terms of Eq. (7) switch on and $R_c$, $H_{qp}$ acquire $V$-dependent values. It is a probe, not a change of the question: the exact poles move only at $O(t_c^2)$ (Sec. 2), so a method that captures the satellites should reproduce them at small $t_c$.

The hopping has to couple $|220\rangle$ to other states, which fixes its form. A hopping $t_c(c_c^\dagger c_o + \text{h.c.})$ is useless: both orbitals are full in $|220\rangle$, so it annihilates the state, and $|220\rangle$ remains an exact eigenstate for every $t_c$ (checked by ED). With $c_v^\dagger c_c$, instead, a core electron moves to the empty $v$ at an energy cost $E_c + g$, so the ground state acquires a core-hole component of amplitude $\sim t_c/(E_c + g)$ and $2 - \nu_c = O(t_c^2)$. This is exactly the combination that Eq. (7) needs to become $V$-dependent: the term $4t_c\, R_c^T \rho^{cv} R_v$ determines $R_c$, and $2 - \nu_c \neq 0$ activates the $o$–$v$ terms.

**Variational problem.**

$$\min_{\{\Phi_I\},\ \psi_{qp}} E \qquad \text{subject to} \qquad \langle \Phi_I | \Phi_I \rangle = 1, \qquad \rho^{II} = {}^I\Delta(\Phi_I) \quad (I = c, o, v). \tag{9}$$

The second constraint ties the local quasiparticle density matrix to the embedding state; it has $B_I(B_I+1)/2$ independent components per fragment (8 for $B_c = 3$). Multipliers $\lambda_I$ of this constraint define the quasiparticle Hamiltonian $H_{qp}$ of Sec. 4.

# 4 Direct minimization

Problem (9) is solved directly (`not_imp_gga_cederbaum.py`), as in the direct-minimization solver used for the Hubbard dimer, without any self-consistency cycle.

**Variables.** The embedding states are parametrized by unnormalized real vectors $x_I$ in the half-filled sector ($\Phi_I = x_I/|x_I|$; dimensions 36, 4, 4 for $B_c = 3$ and 400, 4, 4 for $B_c = 5$), and the Slater determinant by a real $(B_c + 2) \times N_{\mathrm{occ}}$ matrix $X$ with $\rho = X(X^TX)^{-1}X^T$. The constraints $\rho^{II} = {}^I\Delta(\Phi_I)$ are imposed as equalities (SLSQP, tolerance $10^{-13}$ on the energy). Everything is evaluated with sector-projected dense matrices of the operators defined in Sec. 3.

**Gradients.** The analytic gradient of Eq. (7) follows from the chain rule through $R_I(D_I, \Delta_I^{bb})$, $\bar{\Theta}(\nu_c, K, {}^c\Delta)$ and $\rho(X)$:

$$dE = \sum_I 4\, h_I{\cdot}dD_I + \sum_I 2\,\mathrm{Tr}[\Lambda_I\, d\Delta_I^{bb}] + \beta\, d\nu_c + \mathrm{Tr}[\Xi\, dK] + \sum_I dE_I^{loc} + 2\,\mathrm{Tr}[H_{\mathrm{off}}\, d\rho], \tag{10}$$

with $h_I = S_I^{-1}X_I$ and $X_I = \frac{1}{4}\,\partial E/\partial R_I$ given by

$$\begin{aligned} X_c &= t_c\,\rho^{cv}R_v, \qquad X_o = -V\big[(2-\nu_c)\rho^{ov}R_v + \rho^{oc}\bar{\Theta}u\big], \\ X_v &= t_c\,\rho^{vc}R_c - V\big[(2-\nu_c)\rho^{vo}R_o + \rho^{vc}\bar{\Theta}w\big], \end{aligned} \tag{11}$$

The matrices $\Lambda_I = -\frac{1}{2}\nabla^s_{{}^I\Delta}E$ (derivative of $E$ with respect to ${}^I\Delta$ at fixed $D_I$, through $R_I$, $S_I$ and $\bar{\Theta}$) are

$$\begin{aligned} \Lambda_{o,v} &= G\big({}^I\Delta,\, h_I R_I^T\big), \\ \Lambda_c &= G\big({}^c\Delta,\, h_c R_c^T - \tfrac{1}{4}Y\big) - V\nu_c\, C\,(uw^T + wu^T)\,C, \\ Y &= V\big(S_c^{-1}K\,\Pi + \Pi\,K\,S_c^{-1}\big), \quad \Pi = xy^T + yx^T, \quad x = S_c^{-1}w,\ y = S_c^{-1}u, \end{aligned} \tag{12}$$

where $G(\Delta, M) = W\,[(\tilde{M} + \tilde{M}^T) \circ F]\,W^T$ is twice the symmetrized gradient of $\mathrm{Tr}[M\,S(\Delta)]$, $\Delta = W\,\mathrm{diag}(n_a)W^T$, $\tilde{M} = W^TMW$, $F_{ab} = (1 - n_a - n_b)/(s_a + s_b)$, $s_a = \sqrt{n_a(1-n_a) + \varepsilon}$ and $\circ$ is the elementwise product. The remaining derivatives are

$$\beta = \frac{\partial E}{\partial \nu_c} = 4V\big(R_v^T\rho^{vo}R_o + w^TCu\big), \qquad \Xi = \nabla_K E = -V\,\Pi, \tag{13}$$

and $H_{\mathrm{off}}$ has the off-diagonal blocks $H^{co} = -V(\bar{\Theta}u)R_o^T$, $H^{cv} = t_cR_cR_v^T - V(\bar{\Theta}w)R_v^T$, $H^{ov} = -V(2-\nu_c)R_oR_v^T$ (and their transposes), i.e. $\frac{1}{4}\partial E/\partial\rho^{IJ} = H^{IJ}$. The Jacobian of the constraints is also analytic. Gradients and Jacobian are verified against finite differences in Sec. 5.

**Lagrangian and its differential.** With the multipliers $\lambda_I$ (symmetric $B_I \times B_I$, one per independent component of the constraint $\rho^{II} = {}^I\Delta(\Phi_I)$) the Lagrangian of problem (9) is

$$\mathcal{L} = E\big(\{\Phi_I\}, \rho\big) + 2\sum_I \mathrm{Tr}\big[\lambda_I\big({}^I\Delta(\Phi_I) - \rho^{II}\big)\big], \tag{14}$$

where the normalization of $\Phi_I$ and the idempotency of $\rho$ carry no multipliers: they are built into the variables, $\Phi_I = x_I/|x_I|$ and $\rho = X(X^TX)^{-1}X^T$. Since ${}^I\Delta = \mathbb{I} - \Delta_I^{bb}$, the constraint term has the differential $-2\mathrm{Tr}[\lambda_I(d\rho^{II} + d\Delta_I^{bb})]$. Adding it to Eq. (10), the terms in $d\Delta_I^{bb}$ combine into $2\mathrm{Tr}[(\Lambda_I - \lambda_I)\,d\Delta_I^{bb}]$ and the terms in $d\rho$ into $2\mathrm{Tr}[(H_{\mathrm{off}} - \mathrm{blockdiag}(\lambda_I))\,d\rho]$:

$$d\mathcal{L} = \underbrace{\sum_I 4\, h_I{\cdot}dD_I + \sum_I 2\,\mathrm{Tr}[\lambda_I^c\, d\Delta_I^{bb}] + \beta\, d\nu_c + \mathrm{Tr}[\Xi\, dK] + \sum_I dE_I^{loc}}_{\text{variation of the } \Phi_I} + \underbrace{2\,\mathrm{Tr}[H_{qp}\, d\rho]}_{\text{variation of } X}, \tag{15}$$

with the two combinations

$$\lambda_I^c = \Lambda_I - \lambda_I, \qquad H_{qp} = H_{\text{off}} - \text{blockdiag}(\lambda_I). \tag{16}$$

These are the bath potential and the quasiparticle Hamiltonian of the self-consistency loop; here they appear as the coefficients of $d\mathcal{L}$. Stationarity of $\mathcal{L}$ is $d\mathcal{L} = 0$ for every variation of the $\Phi_I$ and of $\rho$ within the projectors:

- *Variation of $X$.* $2\text{Tr}[H_{qp}\, d\rho] = 0$ for all variations of the projector, i.e. $[H_{qp}, \rho] = 0$: $\rho$ is built from eigenvectors of its own $H_{qp}$.

- *Variation of $\Phi_I$.* $D_I$, $\Delta_I^{bb}$, $\nu_c$, $K$ and $E_I^{loc}$ are expectation values in $\Phi_I$ (Sec. 3), so the first group of Eq. (15) is the differential of $\langle\Phi_I|H_I^{imp}|\Phi_I\rangle$ at fixed coefficients, i.e. $2\langle d\Phi_I|H_I^{imp}|\Phi_I\rangle$, with

$$\begin{aligned} H_I^{imp} = H_I^{loc} &+ \sum_a h_{I,a} \sum_\sigma \left(d_{Ia\sigma}^\dagger c_{I\sigma} + c_{I\sigma}^\dagger d_{Ia\sigma}\right) + \sum_{ab} \lambda_{I,ab}^c \sum_\sigma d_{Ia\sigma}^\dagger d_{Ib\sigma} \\ &+ \delta_{Ic}\Big[(\alpha + 2\text{Tr}\Xi)\, n_c - \sum_{\gamma\delta} \Xi_{\gamma\delta}\, n_c \sum_\sigma d_{c\gamma\sigma}^\dagger d_{c\delta\sigma}\Big], \end{aligned} \tag{17}$$

  where $n_c = \sum_\sigma c_{c\sigma}^\dagger c_{c\sigma}$ and $H_c^{loc} = -E_c n_c$, $H_o^{loc} = -g\, n_o$, $H_v^{loc} = +g\, n_v$. The coefficients are

$$h_I = S_I^{-1} X_I, \qquad \lambda_I^c = \Lambda_I - \lambda_I, \qquad \alpha = \beta = 4V\left(R_v^T \rho^{vo} R_o + w^T C u\right), \qquad \Xi = -V\,\Pi, \tag{18}$$

  with $X_I$, $\Lambda_I$, $\Pi$, $\beta$ as in Eq. (10) and below. Each term is the coefficient of one expectation value in the first group of Eq. (15): $h_{I,a}$ multiplies $\langle\sum_\sigma (d_{Ia\sigma}^\dagger c_{I\sigma} + \text{h.c.})\rangle = 4D_{I,a}$; $\lambda_{I,ab}^c$ multiplies $\langle\sum_\sigma d_{Ia\sigma}^\dagger d_{Ib\sigma}\rangle = 2\Delta_{I,ab}^{bb}$; $\alpha + 2\text{Tr}\Xi$ multiplies $\nu_c$; and $-\Xi_{\gamma\delta}$ multiplies $\langle n_c \sum_\sigma d_{c\gamma\sigma}^\dagger d_{c\delta\sigma}\rangle = 2\nu_c\delta_{\gamma\delta} - K_{\gamma\delta}$, which with the $2\text{Tr}\Xi\,\nu_c$ part reproduces $\text{Tr}[\Xi\, dK]$. A constant from normal ordering, $-2\text{Tr}\lambda_I^c$, is dropped. Its vanishing for all $d\Phi_I \perp \Phi_I$ means that $\Phi_I$ is an eigenstate of $H_I^{imp}$. **Note on the excited state of $\Phi_c$.** $\Phi_c$ is not the lowest eigenstate: for $B_c = 3$, $t_c/g = 0.5$ the direct-minimization $\Phi_c$ is the fourth ($V/g = 0.707$) and the sixth ($V/g = 1.414$) of the 36 eigenstates of $H_c^{imp}$ in energy order, with the lowest eigenvalue 2.3 and 8.1 below it, whereas $\Phi_o$ and $\Phi_v$ are ground states (`excited_phi_cederbaum.tex`, `excited_check_golze.py`). The impurity loop, which takes the lowest state, cannot follow it.

The self-consistency loop enforces these conditions one after the other (diagonalizing $H_{qp}$, solving for $\Phi_I$); the direct minimization does not diagonalize either. It passes $E$, $dE$ (Eq. (10)) and the constraints $\kappa = \rho^{II} - {}^I\Delta(\Phi_I)$ with their Jacobian ($d\kappa = d\rho^{II} + d\Delta_I^{bb}$) to SLSQP, which drives the gradient to a constrained stationary point. The multipliers are recovered afterwards from the first group of Eq. (15): stationarity in $\Phi_I$ of $E + \sum_k \mu_k\kappa_k$ is solved for $\mu$ in the least-squares sense (below). The eigenstate $\Phi_I$ is not required to be the lowest one of $H_I^{imp}$.

**Note.** In standard gGA, $H_I^{imp}$ contains only $H_I^{loc}$, the hybridization $h_I$ and the bath term $\lambda_I^c$, and the other fragments enter through $h_I$ alone (the $t_c$ hopping does this). The conditioned $o$–$v$ hopping multiplies a bilinear by $n_c$, so its average also depends on $\nu_c$ and $K$ (through $\bar{\Theta}$, from `main.tex`, `SI.tex`). This adds the $n_c$ and $n_c d^\dagger d$ terms of $H_c^{imp}$ and makes its coefficients $\alpha$, $\Xi$, $\lambda_c^c$ depend on $o$ and $v$ (through $u$, $w$, $\rho^{vo}$), and $h_o$, $h_v$ on the core.

**Starting points.** The problem is non-convex, so several random starts are run per parameter set (embedding states from random small Hamiltonians, $X$ from a random symmetric matrix); a point is accepted if the constraint residual is $< 10^{-9}$, and the lowest accepted energy is kept. Seeding from the previous $t_c$ can end in a poor local minimum (Sec. 6.1), so points are judged by their residuals and not by the energy alone.

$B_c = 5$**.** Random starts rarely converge at $B_c = 5$ (436 variables), so the search is seeded from a $B_c = 3$ solution with the two new ghosts asleep: $\Phi_c^{(5)} = \Phi_c^{(3)} \otimes |\text{ghost}\rangle$ (one extra bath orbital filled, one empty) and $\rho^{(5)} = \rho^{(3)} \oplus \text{diag}(0, 1)$. This point is feasible and has the $B_c = 3$ energy; it is perturbed randomly and minimized. Sparse operators and caching make a $B_c = 5$ run take about 2 s.

**Quasiparticle Hamiltonian and spectrum.** The multipliers are recovered from the stationarity of $E + \sum_k \mu_k \kappa_k$ in $\Phi$ (least squares over the constraint components $\kappa_k = \rho^{II}_{ab} - {}^I\Delta_{ab}$; the residual of this fit is reported as the KKT residual), with $\lambda_{aa} = -\mu_{aa}/2$, $\lambda_{a<b} = -\mu_{ab}/4$. The quasiparticle Hamiltonian is $H_{qp} = H_{\text{off}} - \text{blockdiag}(\lambda_I)$, whose commutator with $\rho$ vanishes at a solution (reported as $\|[H_{qp}, \rho]\|$). The hole Green's function of the core orbital is

$$G^h_{cc}(\omega) = \sum_{n \le N_{\text{occ}}} \frac{|R_c^T \phi_n^{(c)}|^2}{\omega - \epsilon_n - i\eta}, \tag{19}$$

with $\epsilon_n$, $\phi_n$ the eigenvalues (ascending) and eigenvectors of $H_{qp}$ and $\phi_n^{(c)}$ the core block of $\phi_n$; the poles $\epsilon_n$ are compared with $-E_i$ and the weights with $p_i$ [Eq. (3)]. The total weight of $G_{cc}$ is $R_c^T R_c$. The constraint Jacobian is rank deficient at solutions with pure ghost occupations (eigenvalues 0 or 1 of ${}^c\Delta$): some multiplier components are then undetermined (Sec. 6.2).

## 5 Checks

The analytic gradients of Sec. 4 agree with central finite differences of Eq. (7) at random points, for $B_c = 1, 3, 5$, to $\le 4 \times 10^{-10}$ for each identity ($H^{IJ}$, $S_I h_I$, $G(\Delta, M)$, the ${}^c\Delta$ gradient at fixed $R_c$, $\beta = \partial E/\partial \nu_c$, $\Xi = \nabla_K E$), and the complete gradient of the minimization to a relative error of $4 \times 10^{-7}$ (limited by the stiffness of $S^{-1}$ for nearly pure ${}^I\Delta$); the constraint Jacobian to $3 \times 10^{-10}$. The limits are reproduced: at $V = 0$, $E - E_{\text{ref}} = +4.4 \times 10^{-8}$ for $B_c = 1$ and 3 (the effect of $\varepsilon = 10^{-9}$; $\nu_c = 1.995893$ and 1.999835 at $t_c/g = 0.5$ and 0.1); at $t_c = 0$ the energy is exact ($< 10^{-15}$ at $B_c = 1$; $10^{-13}$ and $10^{-11}$ at $B_c = 3$ for $V/g = 0.707$ and 1.414) and $\nu_c = 2$. At $t_c = 0$, $R_c$ is not determined by the stationarity conditions and $V$ does not enter (Sec. 3). The accepted $B_c = 3$ and 5 points have constraint residual $\lesssim 10^{-9}$ (typically $10^{-14}$) and KKT residual $\le 2.5 \times 10^{-6}$.

## 6 Results for $B_c = 1$, 3 and 5

$B_c = 3$ and 5 mean two and four ghost orbitals on the core, $B_c = 1$ none (standard Gutzwiller). $B_c = 3$ and 5 are from the direct minimization ($B_c = 5$ seeded from the $B_c = 3$ solution, plus random starts); $B_c = 1$ from the converged loop, which the direct minimization reproduces to $\le 1.5 \times 10^{-14}$ in $E$ at $t_c/g = 0.1$. All solutions used in the figures and tables are saved (`cederbaum_data/ghost_solutions.npz`).

### 6.1 Energy and $\nu_c$

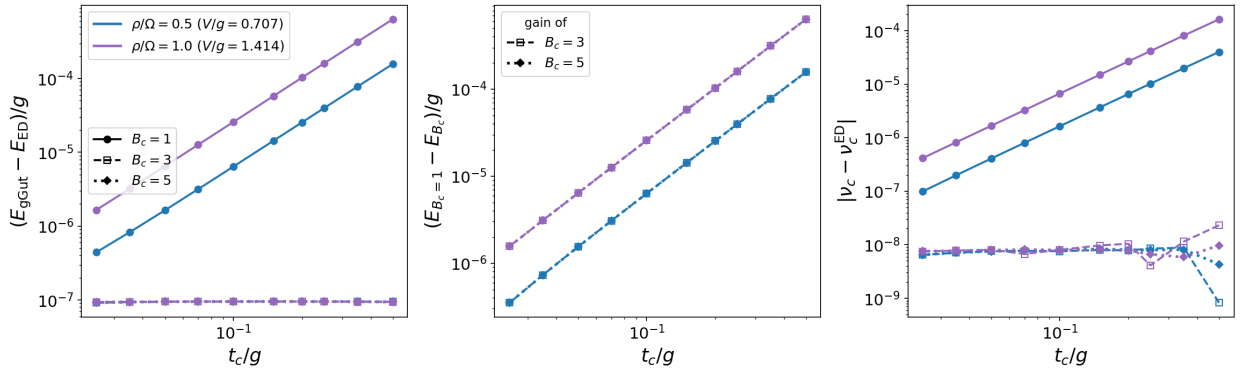



Figure 1: Left: $E_{\mathrm{gGut}} - E_{\mathrm{ED}}$ for $B_c = 1$ (full lines), $B_c = 3$ (dashed, open squares) and $B_c = 5$ (dotted, diamonds); blue: $\rho/\Omega = 0.5$ ($V/g = 0.707$), purple: $\rho/\Omega = 1.0$ ($V/g = 1.414$). Middle: energy gained by the ghosts, $E_{B_c=1} - E_{B_c}$ (the two coincide). Right: $|\nu_c - \nu_c^{\mathrm{ED}}|$. Log–log axes; $E_c/g = 10$.

- $B_c = 1$**:** $E_{\mathrm{gGut}} - E_{\mathrm{ED}} \simeq 1.3 \times 10^{-3}\, V^2 t_c^2$ over $t_c/g \in [0.025, 0.5]$ (exponent 1.97 and 1.99), and $|\nu_c - \nu_c^{\mathrm{ED}}| \propto t_c^2$.
- $B_c = 3$**:** $E_{\mathrm{gGut}} - E_{\mathrm{ED}} = 9.2$–$9.6 \times 10^{-8}$ for all $(V, t_c)$, flat; $|\nu_c - \nu_c^{\mathrm{ED}}| \le 2.3 \times 10^{-8}$. This floor is the regularization shift, not the ansatz.
- $B_c = 5$**:** the same energy at all 20 points ($E_{B_c=5} - E_{B_c=3}$ between $-6 \times 10^{-11}$ and $+6 \times 10^{-13}$; KKT $\le 2.3 \times 10^{-6}$); 5–9 of 14 runs per point converged to the same energy, and 3 of 120 random starts. The extra ghosts do not lower the energy.

The ghosts thus recover 79–99.99% of the $B_c = 1$ error (the smallest fractions at the smallest $t_c$, where the shift floor is comparable to the error). Poor local minima from continuation ($(V/g, t_c/g) = (0.707, 0.025)$, $(1.414, 0.07)$) were recomputed from random starts; one $B_c = 1$ loop point needed mixing 0.3; two $B_c = 5$ seeds were obtained by continuation.

### 6.2 Hole spectral function

| | | ED main pole | | gGut $B_c = 1$ | | gGut $B_c = 3$ | | gGut $B_c = 5$ | |
|---|---|---|---|---|---|---|---|---|---|
| $V/g$ | $t_c/g$ | position | weight | position | weight | position | weight | position | weight |
| 0.707 | 0.1 | −9.5516 | 0.8250 | −10.0007 | 0.9999 | −9.9355 | 0.9940 | −9.9355 | 0.9940 |
| 0.707 | 0.05 | −9.5508 | 0.8249 | −9.9997 | 0.9999 | −9.9330 | 0.9939 | −9.9330 | 0.9939 |
| 1.414 | 0.1 | −8.5373 | 0.6222 | −10.0005 | 0.9999 | −9.7458 | 0.9767 | −9.7458 | 0.9767 |
| 1.414 | 0.05 | −8.5362 | 0.6221 | −9.9997 | 0.9999 | −9.7435 | 0.9766 | −9.7434 | 0.9766 |

Table 3: Dominant hole pole of $G_{cc}$. ED additionally has satellites at about −12.0 (total weight 0.167 and 0.333 for $V/g = 0.707$, 1.414) and at −14.45 and −15.47 (weights 0.008 and 0.045); gGut has none: all other occupied levels have weight $\le 1.5 \times 10^{-6}$ ($B_c = 1$) and $\le 2.2 \times 10^{-6}$ ($B_c = 3$, 5).

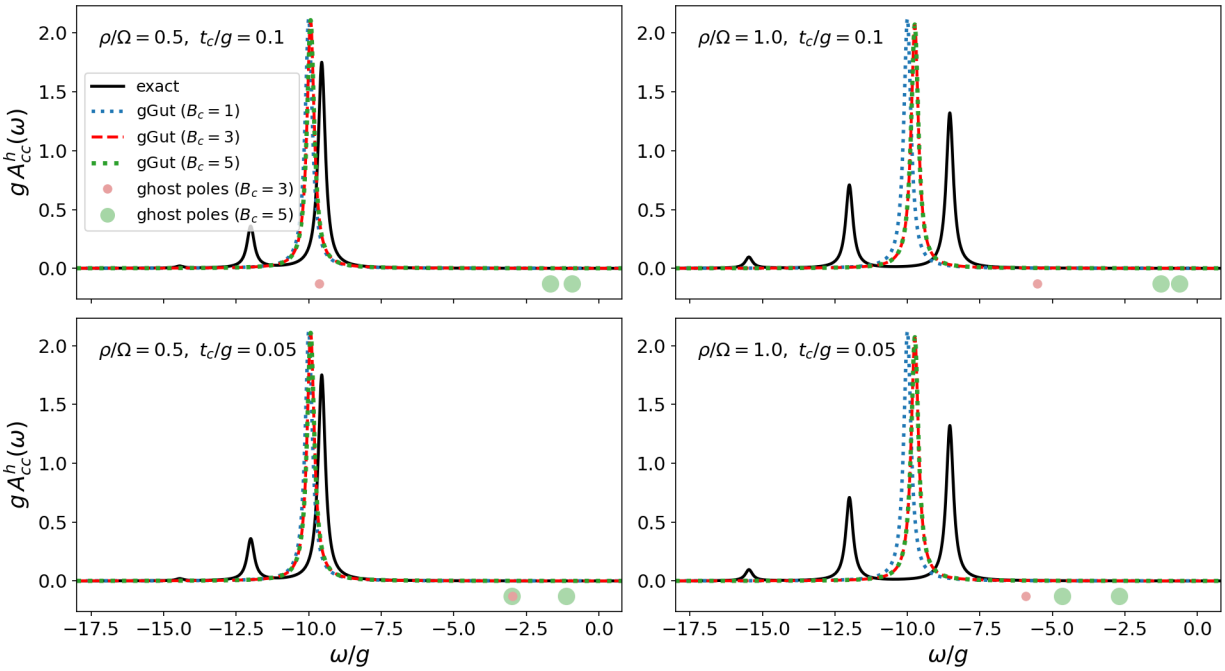



Figure 2: Hole spectral function $A^h_{cc}(\omega) = \sum_n w_n\,(\eta/\pi)/[(\omega-\omega_n)^2+\eta^2]$, $\eta = 0.15$: exact (solid), gGut $B_c = 1$ (blue dotted), $B_c = 3$ (red dashed), $B_c = 5$ (green dotted, on top of $B_c = 3$). Rows: $t_c/g = 0.1$, 0.05; columns: $\rho/\Omega = 0.5$ ($V/g = 0.707$), 1.0 ($V/g = 1.414$); $E_c/g = 10$. Dots below zero: the occupied muted poles (zero-weight ghost levels of $H_{qp}$, defined in the text), as returned by the minimizer: one at $B_c = 3$ (small, rose), two at $B_c = 5$ (large, green); $B_c = 1$ has none. Their positions are not fixed by the solution (see text) and they carry no weight.

**Result.** The $B_c = 1$ pole sits at $-10.000$ for every $V$: the bare core level. At $B_c = 3$ and 5 (identical to $10^{-4}$) the single pole moves with $V$ towards the exact main pole but covers only 15–18% of the exact displacement from $-E_c$ (0.067 of 0.449 and 0.257 of 1.464 at $t_c/g = 0.05$), and its weight (0.994, 0.977) stays far above the exact one (0.825, 0.622). *No satellites appear*; the ghosts only add $V$ dependence to the main pole.

**Why there is no weight on the ghosts (interpretation).** The weights in Eq. (19) are $|R_c^T \phi_n^{(c)}|^2$. At $B_c = 3$ ($t_c/g = 0.1$) the natural orbitals of ${}^c\Delta$ have occupations $\sim 10^{-7}$, 0.9999 and 1 (the nearly full core orbital and two pure ghosts), and $R_c$ lies entirely along the core orbital (components along the others $\leq 10^{-4}$), so the occupied ghost level carries $3\times10^{-7}$ and $4\times10^{-10}$ at $V/g = 0.707$, 1.414 ($\leq 2.3\times10^{-9}$ at $B_c = 5$), and $R_c^T R_c = 0.994$, 0.977 sits on one pole. Mixing the two filled core orbitals by hand in $\lambda_c$ does split the weight (e.g. 0.57/0.42 at $V/g = 0.707$, $\delta = 1$), but $\|[H_{qp},\rho]\|$ then grows from $\leq 2.6\times10^{-5}$ to $10^{-2}$–$4\times10^{-2}$: these points are not stationary, and their poles and weights do not match the exact ones. The ghosts appear to carry the correlation through $\nu_c$ and $K$ (the terms $-V(\bar\Theta u)R_o^T$, $-V(\bar\Theta w)R_v^T$ of $H_{qp}$) but not through $R_c$, which is all that Eq. (19) sees.

**Muted poles.** In gGut every pole of the spectral function is an eigenvalue of $H_{qp}$, which has $B_c + B_o + B_v$ levels. Three of them are the quasiparticle (QP) levels of $c$, $o$ and $v$, the levels where the orbital $c$, $o$ or $v$ has its largest weight: the core hole near $-10$, the occupied level near $-g$ and

the virtual level near $+g$. The remaining $B_c+B_o+B_v-3$ levels are the *muted poles*: two at $B_c=3$ (with $B_o=B_v=1$) and four at $B_c=5$. Occupied ones are hole poles, empty ones particle poles. Their weight $|R_I^T\phi_n^{(I)}|^2$ is $\leq 2.2\times10^{-6}$ on every orbital, so they are invisible in $A(\omega)$. They matter because they are the only place where gGut could put a satellite: a satellite needs a pole of $H_{qp}$ at $\omega\simeq-12$ or lower, with weight.

- *Character.* The hole-side ones are ghost levels of the core block ($>0.99$), with core weight $<10^{-4}$. They are counted by occupation, not by energy order, since a free ghost level can cross the level $o$ (in the $B_c=5$ solution at $t_c/g=0.05$, $V/g=1.414$ the $o$ level is empty and a ghost level is occupied). The Fig. 2 dots are the occupied ones: one at $B_c=3$, two at $B_c=5$. The empty level at $\omega\simeq+12$ (Fig. 3) is not a ghost: its weight sits on $o$ ($\simeq10^{-7}$–$10^{-6}$) and it matches the exact $o$ particle peak at 12.003.

- *Position.* It is not fixed by the solution: the constraint Jacobian has rank 7 of 8 ($B_c=3$) and 13 of 17 ($B_c=5$), and shifting the multipliers along the undetermined directions moves the ghost levels, for example to the exact satellite energies, without changing $E$, the main pole, its weight or $\|[H_{qp},\rho]\|$; their weights stay $\leq 8\times10^{-9}$. Different minimizer solutions of the same point place them differently ($-2.98$ and $-5.93$ in the figures, $-2.20$ and $-9.53$ in earlier solutions, at $B_c=3$, $t_c/g=0.05$); the returned values are the minimum-norm multipliers. They lie between the core level ($\approx-10$) and the valence region ($\approx-1$), because the ghost levels come from the multipliers of the density constraint, which are of valence-scale magnitude (the bath levels of $\lambda_c^c$ at $t_c/g=0.5$ are $-2.1$ to $+1.7$ and $-5.1$ to $+3.0$; `excited_phi_cederbaum.tex`), and not from $E_c$.

- *Weight.* It is set by $R_c$, which the energy minimization fixes along the core orbital. A solution with weight on a muted pole would have a different $R_c$ and so differ from the minimum; none was found.

- *Ghosts on o and v.* With $B_c=B_o=B_v=3$ (9 levels, 6 muted poles) I seeded the minimization from the $B_c=3$ solution plus a sleeping ghost pair on $o$ and on $v$ (same energy), perturbed it (15 runs per case), and also tried random starts. No run went below the seed energy; the random starts mostly fell to the trivial product state ($R_c=0$). In three of the four cases of Fig. 3 a run returns to the $B_c=3$ point ($E-E_{\rm ED}\simeq9.5\times10^{-8}$, same main pole and weight); at $V/g=0.707$, $t_c/g=0.1$ the best point found is $B_c=1$-like ($6.4\times10^{-6}$). The extra poles again carry no weight (purple row of Fig. 3). The search is limited (SLSQP, seeds from $B_c=3$ only).

The exact satellites lie *below* the main peak ($-12$ and $-14.5$ at $V/g=0.707$, $-15.5$ at $V/g=1.414$, against $-9.6$ exact, $-9.9$ gGut), whereas the muted poles lie above it, so none of them can play the role of a satellite.

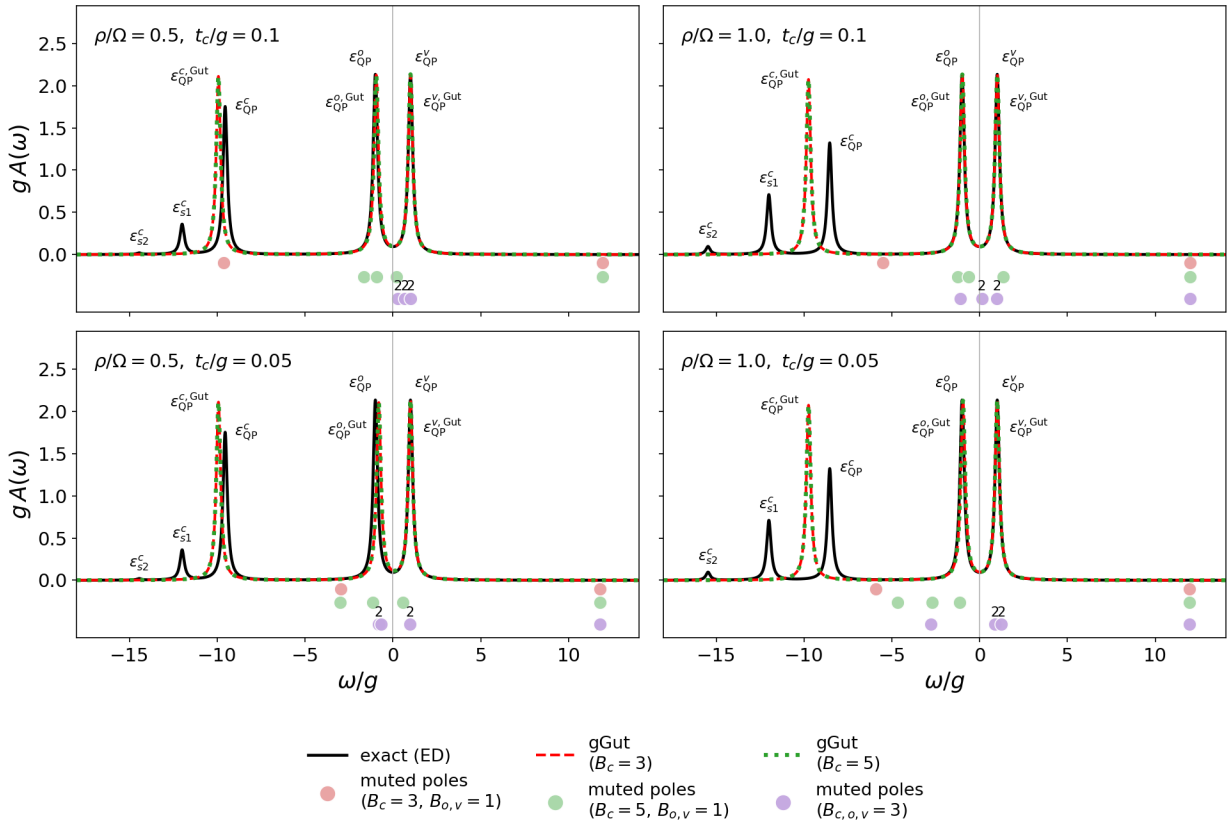



Figure 3: Full spectral function $A(\omega) = \sum_{I=c,o,v} A_{II}(\omega)$ (hole side $\omega < 0$, particle side $\omega > 0$; per spin, $\eta = 0.15$, $E_c/g = 10$). Rows: $t_c/g = 0.1$, 0.05; columns: $\rho/\Omega = 0.5$, 1.0. *Curves:* exact (black), gGut $B_c = 3$ (red dashed) and $B_c = 5$ (green dotted; both with $B_o = B_v = 1$, from the cached solutions of Fig. 2). *Labels:* $\varepsilon^I_{\mathrm{QP}}$ is the quasiparticle peak of the exact spectrum for the orbital $I = c, o, v$ and $\varepsilon^c_{s1}$, $\varepsilon^c_{s2}$ are the exact core satellites, numbered by increasing distance from $\varepsilon^c_{\mathrm{QP}}$; $\varepsilon^{I,\mathrm{Gut}}_{\mathrm{QP}}$ are the gGut quasiparticle peaks (the core one is single, with no satellites). *Dots below the axis:* the muted poles of gGut, i.e. all eigenvalues of $H_{qp}$ except the three quasiparticle levels. One row per solution: $B_c = 3$ (rose, 2 poles), $B_c = 5$ (green, 4) and $B_c = B_o = B_v = 3$ (purple, 6; the lowest-energy run seeded from the $B_c = 3$ solution with sleeping ghosts on $o$ and $v$; in the panel $\rho/\Omega = 0.5$, $t_c/g = 0.1$ no run returned to the seed and the best point is $B_c = 1$-like). Poles closer than 0.02 share one dot, with the multiplicity written above it when it is larger than 1. The muted poles carry no weight ($\leq 2.2 \times 10^{-6}$) and none lies near the exact satellites.

**Full spectral function (core, occupied, virtual).** Fig. 3 shows all peaks of $A = A_{cc} + A_{oo} + A_{vv}$, hole side ($\omega < 0$) and particle side ($\omega > 0$), against ED. In gGut every eigenvalue of $H_{qp}$ is a pole, occupied ones on the hole side and empty ones on the particle side, with weight $|R_I^T \phi_n^{(I)}|^2$ on orbital $I$. The three main peaks are the core hole near $-10$ ($c$), the occupied level near $-g$ ($o$) and the virtual level near $+g$ ($v$); gGut reproduces the last two, and the core pole only approximately (position and weight as in Tab. 3). What gGut misses are the exact core satellites at $-12$ and $-14.5$ ($-15.5$); the remaining ED peaks ($\omega \simeq 3$ for $c$ and $\omega \simeq 12$ for $o$, particle side) have weight $\lesssim 10^{-6}$ and are not visible. The remaining levels are the muted poles defined above ($B_c = 3$: one hole pole at $-5.9$ to $-3.0$ and the empty level at $\simeq +12$; $B_c = 5$: several between $-4.7$ and $+1.4$ besides that one), all with weight $\leq 10^{-6}$.

### 6.3 Comparison with the three-level model of Cederbaum and of Kockläuner and Golze

The orbitals $c$, $o$, $v$ are the three-level core-hole model of Cederbaum, Domcke and Schirmer [1] (Eqs. (28)–(29)), used by Kockläuner and Golze [2] (Fig. 2) to test $G_0W_0$ and GW plus cumulant (GW+C). With $\Omega = \epsilon_v - \epsilon_o = 2g$ and the transition moment $\rho$ (GW+C excitation weight $\gamma = \rho^2/\Omega^2$), the exact solution is the $3\times 3$ matrix with off-diagonal elements $-\sqrt{2}V$, which gives Eq. (3) and the weights $(a \pm \Omega)^2/4a^2$, $2V^2/a^2$ of Ref. [1] ($a = (\Omega^2 + 4V^2)^{1/2}$). The $\sqrt{2}$ is the normalization of the two spin channels, so

$$\rho = \sqrt{2}\, V, \qquad \frac{\rho}{\Omega} = \frac{V}{\sqrt{2}\, g}, \tag{20}$$

confirmed by the small-$V$ limit of the exact satellite-to-QP weight ratio ($2.00\times 10^{-4} = (\sqrt{2}V)^2/\Omega^2$ at $V/g = 0.02$, `g0w0_cederbaum.py`). Their ratios $\rho/\Omega = 0.5$, 1.0 are $V/g = 0.707$, 1.414, the values used throughout this note. Their figure gives only the ratio, no energy axis, so $E_c$ cannot be taken from it (it only shifts the poles rigidly); Fig. 4 therefore uses $\omega$ with ticks at the exact $\epsilon_c^{\mathrm{QP}}$, $\epsilon_c^{\mathrm{Sat1}}$, $\epsilon_c^{\mathrm{Sat2}}$.

**$G_0W_0$ and GW+C for the model (computed here).** We computed $G_0W_0$ and GW+C for this model ourselves (`g0w0_cederbaum.py`) from the expressions below, in order to compare with the $G_0W_0$ and GW+C of Ref. [2] (not taken from it). With $G_0$ the bare core level and one bosonic mode, $\Sigma_c(\omega) = \rho^2/(\omega - \epsilon_c + \Omega)$ (no exchange or Hedin shift), the poles of $[\omega - \epsilon_c - \Sigma_c]^{-1}$ solve $x(x+\Omega) = \rho^2$: a QP pole and one satellite at $-\Omega\sqrt{1 + 4\rho^2/\Omega^2}$, against the exact satellites at $-\Omega\sqrt{1 + 2\rho^2/\Omega^2}$ and twice that. GW+C has Poisson satellites at $\epsilon_{\mathrm{QP}} - n\Omega$ with weights $e^{-\gamma}\gamma^n/n!$. These expressions agree with Eqs. (11), (14), (16)–(17), (33), (36), (44)–(47) of Ref. [2] (the exchange constant $\Sigma^x - v^{xc}$ is absorbed in $\epsilon_c$), and the first-satellite positions measured in their rendered Fig. 2b,c (in units of $\Omega$, accuracy a few per cent) agree with this note: exact 1.22 and 1.76 (here 1.225, 1.732), $G_0W_0$ 1.42 and 2.23 (here 1.414, 2.236) at $\rho/\Omega = 0.5$, 1.0.

| $\rho/\Omega$ | | satellite positions /$\Omega$ (from QP) | weights (QP, satellites) |
|---|---|---|---|
| 0.5 | exact | $-1.225$, $-2.449$ | 0.825, 0.167, 0.008 |
| | $G_0W_0$ | $-1.414$ | 0.854, 0.146 |
| | GW+C | $-1$, $-2$, $-3$, … | 0.779, 0.195, 0.024, … |
| 1.0 | exact | $-1.732$, $-3.464$ | 0.622, 0.333, 0.045 |
| | $G_0W_0$ | $-2.236$ | 0.724, 0.276 |
| | GW+C | $-1$, $-2$, $-3$, … | 0.368, 0.368, 0.184, … |

Table 4: Exact ($t_c = 0$), $G_0W_0$ and GW+C hole spectrum of the model at the ratios of Fig. 2 of Ref. [2] ($V/g = 0.707$, 1.414).

**What matches.** The exact spectrum has the QP peak and two satellites (a single and a double $o \to v$ excitation, the second much weaker), with intensity and QP–satellite separation growing with $\rho/\Omega$; $G_0W_0$ has one satellite at higher binding energy than the exact one ($-1.41\,\Omega$ against $-1.23\,\Omega$ at $\rho/\Omega = 0.5$; $-2.24\,\Omega$ against $-1.73\,\Omega$ at 1.0); GW+C puts satellites at $-n\Omega$ (closer to the exact ones at $\rho/\Omega = 0.5$ than at 1.0) and shows spurious higher satellites at larger $\rho/\Omega$; all as stated in Ref. [2]. The hopping $t_c$ moves the exact poles by at most 0.06 at $t_c/g = 0.5$.

**What does not match.** gGut gives a single pole at both ratios (weight 0.977–0.9999): it reproduces neither the exact satellites nor the GW+C or $G_0W_0$ ones, and does worse than $G_0W_0$ for

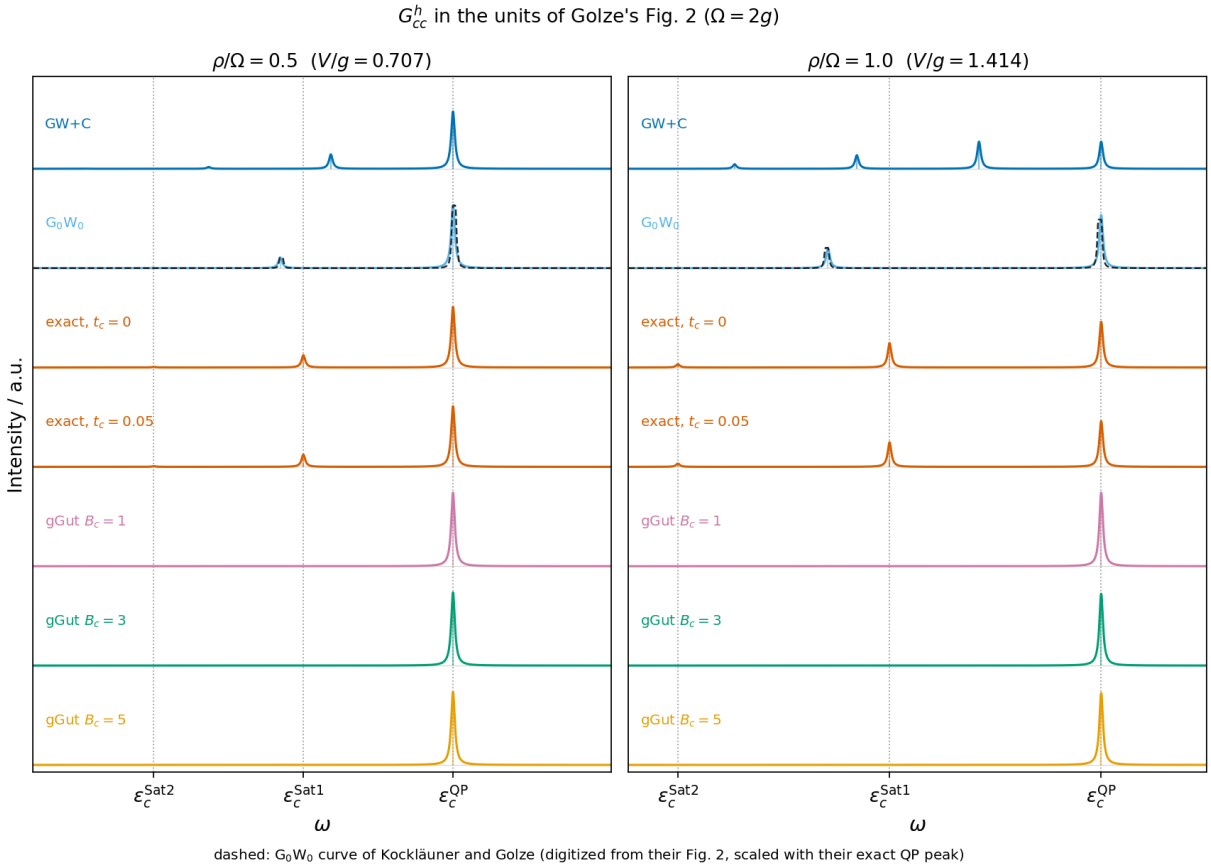



Figure 4: $G^h_{cc}$ in the units of Fig. 2 of Ref. [2]; left $\rho/\Omega = 0.5$ ($V/g = 0.707$), right $\rho/\Omega = 1.0$ ($V/g = 1.414$), $E_c/g = 10$. Rows: GW+C; $G_0W_0$; exact at $t_c = 0$ and 0.05; gGut $B_c = 1$, 3, 5 ($t_c/g = 0.05$). All rows share one scale (peak height = pole weight) and a Lorentzian half-width $0.018\,\Omega$, fitted to the $G_0W_0$ peaks of Golze. Ticks: exact $t_c = 0$ QP and satellite energies (dotted lines). Dashed: the $G_0W_0$ curve of Ref. [2], digitized from their Fig. 2 and scaled with their exact QP peak (`measure_golze_fig2.py`); its QP peak at $\rho/\Omega = 1.0$ is cut off by their frame.

this model, whose QP pole is within 0.04 and 0.23 of the exact one ($-9.59$, $-8.76$ against $-9.55$, $-8.54$) while the gGut $B_c = 3$ pole is at $-9.93$, $-9.74$ (errors 0.38, 1.21) and the $B_c = 1$ pole at $-10.00$. The muted poles (dots of Fig. 2) are not satellites: no weight and arbitrary positions, as discussed above.

# 7 Conclusions

1. **gGut does not reproduce the satellite structure of the model.** For $B_c = 1$, 3, 5, $\rho/\Omega = 0.5$, 1.0 ($V/g = 0.707$, 1.414) and $t_c/g = 0.05$–0.5, the hole spectral function has a single pole with weight 0.977–0.9999, whereas the exact spectrum has satellites near $-12$ and at $-14.45$, $-15.47$ carrying 17.5% and 37.8% of the weight. The main pole is displaced from the exact one ($-9.93$ vs. $-9.55$; $-9.74$ vs. $-8.54$).

2. The ghosts improve the ground state: $E - E_{\mathrm{ED}} \approx 10^{-7}$ at $B_c = 3$ and 5 (vs. $1.3 \times 10^{-3} V^2 t_c^2$ at $B_c = 1$), $\nu_c$ to $\sim 10^{-8}$; $B_c = 5$ returns the $B_c = 3$ energy and spectrum (the extra ghosts

stay asleep).

3. The muted poles (the levels of $H_{qp}$ beyond the three QP levels) have zero weight ($R_c$ lies along the core orbital) and arbitrary energy (free multiplier directions): moving them to the exact satellite energies leaves $E$ and the weights unchanged, so they give no satellite. A solution with satellite weight would have to change $R_c$, i.e. differ from the energy minimum; none was found.

4. The model uses a core–virtual hopping: a core–occupied hopping leaves $|220\rangle$ exact for every $t_c$ and cannot make $R_c$ determined. The energy functional and its gradients are verified (finite differences $\sim 10^{-10}$), and the limits $V = 0$, $t_c = 0$ are reproduced.

5. The model is the three-level model of Cederbaum *et al.* [1] with $\rho = \sqrt{2}V$; for $\rho/\Omega = 0.5$, 1.0 the exact spectrum has the QP peak and two satellites, $G_0W_0$ one satellite at too high a binding energy, GW+C satellites at $-n\Omega$, and gGut a single pole (Fig. 4, Tab. 4).

# A The quasiparticle Hamiltonian as a matrix

$H_{qp}$ acts on the $B_c+2$ quasiparticle orbitals of one spin, ordered as the $B_c$ core orbitals $d_{c1}, \ldots, d_{cB_c}$, then $d_o$, then $d_v$ (the block structure c, o, v of $\rho$ in Sec. 3). It is real and symmetric. Its ingredients, all defined in Sec. 3, are the renormalization vectors $R_c \in \mathbb{R}^{B_c}$, $R_o, R_v \in \mathbb{R}$, the core occupation $\nu_c$, the tensor $\bar{\Theta} \in \mathbb{R}^{B_c \times B_c}$ and the two core vectors

$$a = \bar{\Theta}\, u = \bar{\Theta}\, \rho^{cv} R_v, \qquad b = \bar{\Theta}\, w = \bar{\Theta}\, \rho^{co} R_o \qquad (a, b \in \mathbb{R}^{B_c}). \tag{21}$$

The diagonal blocks are the multipliers of the constraint $\rho^{II} = {}^{I}\Delta$ of Eq. (9), $-\lambda_c \in \mathbb{R}^{B_c \times B_c}$ (symmetric) and $-\lambda_o, -\lambda_v \in \mathbb{R}$; the off-diagonal blocks are $H^{IJ} = \frac{1}{4}\partial E/\partial \rho^{IJ}$ of Sec. 4. In block form, with the blocks of size $B_c$, 1, 1,

$$H_{qp} = \begin{pmatrix} -\lambda_c & -V R_o\, a & R_v\,(t_c R_c - V\, b) \\ -V R_o\, a^T & -\lambda_o & -V(2-\nu_c)\, R_o R_v \\ R_v\,(t_c R_c - V\, b)^T & -V(2-\nu_c)\, R_o R_v & -\lambda_v \end{pmatrix}. \tag{22}$$

The *co* and *cv* blocks are the column vectors $-V R_o\, a$ and $R_v(t_c R_c - V b)$ ($B_c \times 1$); the *ov* block is a number. This is what `build_Hqp` returns (`gga_cederbaum.py`), and it is the same operator as $H_{\text{off}} - \text{blockdiag}(\lambda_I)$ of Sec. 4.

**Spin.** $H_{qp}$ and $\rho$ in this appendix are the *one-spin* matrices, of size $B_c + 2$: $\rho_{pq} = \langle d^\dagger_{p\sigma} d_{q\sigma} \rangle$ with $\sigma$ fixed, and the solutions are paramagnetic, so $\uparrow$ and $\downarrow$ have the same matrix. The quasiparticle Hamiltonian on the full space of $2(B_c + 2)$ spin orbitals, ordered ($\uparrow$ block, $\downarrow$ block), is

$$H^{\text{full}}_{qp} = \begin{pmatrix} H_{qp} & 0 \\ 0 & H_{qp} \end{pmatrix} = \mathbb{1}_2 \otimes H_{qp}, \qquad \rho^{\text{full}} = \mathbb{1}_2 \otimes \rho, \tag{23}$$

and each spin channel fills its $N_{\text{occ}}$ lowest levels ($N_{\text{occ}} = 2$ for $B_c = 1$, 3 for $B_c = 3$), i.e. $2N_{\text{occ}}$ electrons in total. The spin sum and the hermitian conjugate that give the factor 4 in Eq. (7) are removed in $H^{IJ} = \frac{1}{4}\partial E/\partial \rho^{IJ}$, so the entries below carry no spin factor.

**Explicit form for $B_c = 1$.** There are no ghosts, $\lambda_c$, $a$, $b$, $R_c$, $\bar{\Theta}$ are numbers and $H_{qp}$ is $3 \times 3$:

$$H_{qp}^{B_c=1} = \begin{pmatrix} -\lambda_c & -VR_o\, a & R_v(t_c R_c - V\, b) \\ -VR_o\, a & -\lambda_o & -V(2-\nu_c)R_o R_v \\ R_v(t_c R_c - V\, b) & -V(2-\nu_c)R_o R_v & -\lambda_v \end{pmatrix}, \tag{24}$$

with $a = \bar{\Theta} R_v \rho^{cv}$ and $b = \bar{\Theta} R_o \rho^{co}$.

Diagonalizing it, $H_{qp}\phi_n = \epsilon_n \phi_n$ ($\epsilon_1 \le \epsilon_2 \le \epsilon_3$, $\phi_{n,p}$ the component of $\phi_n$ on orbital $p$), and filling the $N_{\mathrm{occ}} = 2$ lowest levels per spin gives $\rho_{pq} = \langle d^\dagger_{p\sigma} d_{q\sigma} \rangle = \sum_{n=1}^{2} \phi_{n,p}\phi_{n,q} \equiv s_{pq}$:

$$\rho^{B_c=1} = \begin{pmatrix} s_{cc} & s_{co} & s_{cv} \\ s_{co} & s_{oo} & s_{ov} \\ s_{cv} & s_{ov} & s_{vv} \end{pmatrix}. \tag{25}$$

The diagonal entries are the new ${}^I\Delta$, and the off-diagonal ones give the $a$, $b$ of the next $H_{qp}$ through $u = R_v s_{cv}$, $w = R_o s_{co}$: $a = \bar{\Theta} R_v s_{cv}$, $b = \bar{\Theta} R_o s_{co}$.

**Explicit form for $B_c = 3$.** With $\lambda_c = (\lambda_{ij})$ symmetric, $a = (a_1, a_2, a_3)^T$, $b = (b_1, b_2, b_3)^T$, $R_c = (R_{c1}, R_{c2}, R_{c3})^T$ and $\tau_i = t_c R_{ci} - V b_i$, $H_{qp}$ is $5 \times 5$:

$$H_{qp}^{B_c=3} = \begin{pmatrix} -\lambda_{11} & -\lambda_{12} & -\lambda_{13} & -VR_o a_1 & R_v\tau_1 \\ -\lambda_{12} & -\lambda_{22} & -\lambda_{23} & -VR_o a_2 & R_v\tau_2 \\ -\lambda_{13} & -\lambda_{23} & -\lambda_{33} & -VR_o a_3 & R_v\tau_3 \\ -VR_o a_1 & -VR_o a_2 & -VR_o a_3 & -\lambda_o & -V(2-\nu_c)R_o R_v \\ R_v\tau_1 & R_v\tau_2 & R_v\tau_3 & -V(2-\nu_c)R_o R_v & -\lambda_v \end{pmatrix}. \tag{26}$$

Diagonalizing it ($\epsilon_1 \le \cdots \le \epsilon_5$) and filling the $N_{\mathrm{occ}} = 3$ lowest levels per spin gives $\rho_{pq} = \sum_{n=1}^{3} \phi_{n,p}\phi_{n,q} \equiv s_{pq}$:

$$\rho^{B_c=3} = \begin{pmatrix} s_{c_1c_1} & s_{c_1c_2} & s_{c_1c_3} & s_{c_1o} & s_{c_1v} \\ s_{c_1c_2} & s_{c_2c_2} & s_{c_2c_3} & s_{c_2o} & s_{c_2v} \\ s_{c_1c_3} & s_{c_2c_3} & s_{c_3c_3} & s_{c_3o} & s_{c_3v} \\ s_{c_1o} & s_{c_2o} & s_{c_3o} & s_{oo} & s_{ov} \\ s_{c_1v} & s_{c_2v} & s_{c_3v} & s_{ov} & s_{vv} \end{pmatrix}. \tag{27}$$

The upper-left $3 \times 3$ block is the new ${}^c\Delta$ and $s_{oo}$, $s_{vv}$ the new ${}^o\Delta$, ${}^v\Delta$. The off-diagonal entries give the $a$, $b$ of the next $H_{qp}$: $u = R_v(s_{c_1v}, s_{c_2v}, s_{c_3v})^T$, $w = R_o(s_{c_1o}, s_{c_2o}, s_{c_3o})^T$, $a = \bar{\Theta} u$, $b = \bar{\Theta} w$.

For $B_c = 5$ the core block is $5 \times 5$ and the two border columns have five entries each, same pattern.

**Use in the impurity loop.** In the impurity loop (`gga_cederbaum.py`) $H_{qp}$ is diagonalized once per macroiteration; there is no inner loop on $\rho$. The $\rho$ obtained from the occupied eigenvectors is used directly in the next macroiteration: its diagonal blocks $\rho^{II}$ are the ${}^I\Delta$ that enter the embedding problems (before the linear mixing with their output), and its off-diagonal blocks give $u$, $w$ and hence $a$, $b$ in the next $H_{qp}$. $\rho$ is thus self-consistent with $H_{qp}$ only at convergence of the macroiterations. (Repeating the diagonalization until $\rho$ stops changing, before the embedding step, was tested and does not change the converged result for $B_c = 1$ nor make the $B_c = 3$ loop converge.)

In the direct minimization (Sec. 4) no such cycle is needed: $\rho = X(X^TX)^{-1}X^T$ is a variable of the minimization, $E$ already contains the dependence of $H_{qp}$ on $\rho$ through $u$ and $w$, and the

condition that $\rho$ is built from its own $H_{qp}$ is the stationarity condition $[H_{qp}, \rho] = 0$; the link $\rho^{II} = {}^{I}\Delta(\Phi_I)$ is an equality constraint. At the minimizers found, building $H_{qp}$ from the multipliers and from this $\rho$, diagonalizing it and filling the $N_{\mathrm{occ}}$ lowest levels returns the same $\rho$ to the accuracy of the stationarity ($\|\rho_{\mathrm{filled}} - \rho\| = 10^{-8}$ for $B_c = 1$ and for $B_c = 3$ at $\varepsilon = 10^{-5}$; $2 \times 10^{-5}$ for $B_c = 3$ at $\varepsilon = 10^{-9}$, where $\|[H_{qp}, \rho]\| = 3 \times 10^{-4}$), and the occupied levels are the lowest ones, with a gap 1–2 to the first empty level ($V/g = 1$, $t_c/g = 0.5$).

# References


[1] L. S. Cederbaum, W. Domcke and J. Schirmer, Many-body theory of core holes, Phys. Rev. A **22**, 206 (1980).

[2] J. Kockläuner and D. Golze, GW plus cumulant approach for predicting core-level shake-up satellites in large molecules, arXiv:2412.17132 (2024).